

- 6 (a) A uniform wire AB of length 150 cm and radius 0.45 mm is connected in series with a cell E of e.m.f. 6.0 V with negligible internal resistance as shown in Fig. 6.1.

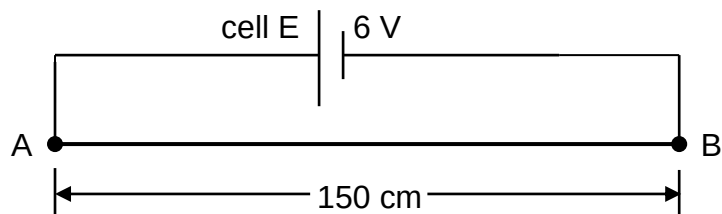


Fig. 6.1

The resistivity of the wire is $1.1 \times 10^{-6} \Omega \text{ m}$.

Determine the resistance of the wire.

resistance = Ω [2]

- (a) **Solution:**
The resistance of the wire

$$\begin{aligned}
 &= \frac{\rho L}{A} \\
 &= \frac{(1.1 \times 10^{-6})(150 \times 10^{-2})}{\pi (0.45 \times 10^{-3})^2} \\
 &= 2.59 \Omega
 \end{aligned}$$

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- (b)** A cell C of e.m.f. 1.5 V with internal resistance r is connected to the circuit of Fig. 6.1, as shown in Fig. 6.2.

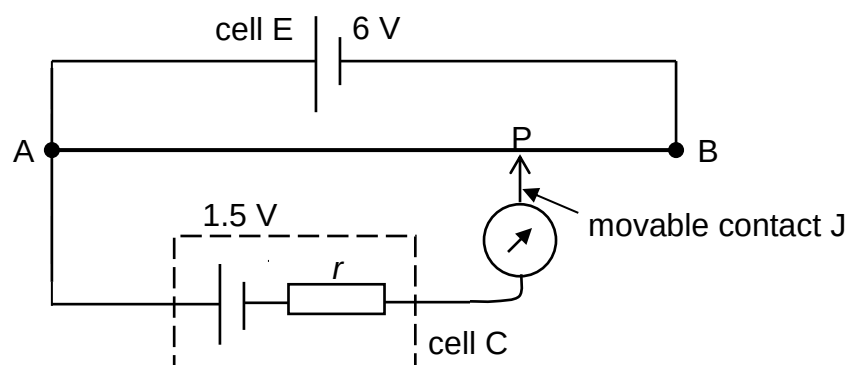


Fig. 6.2

The movable contact J can be connected to any point along the wire AB.

- (i)** Determine the length AP required to produce zero current in the galvanometer.

length AP = m [2]

- (i) Solution:**

$$\frac{V_C}{V_E} = \frac{l_{AP}}{150}$$

$$\frac{1.5}{6} = \frac{l_{AP}}{150}$$

$$l_{AP} = 0.375 \text{ m}$$

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- (ii) An additional resistor, R , of resistance $2.0\ \Omega$ is now connected between the terminals of cell C as shown in Fig. 6.3, and it produces a balance length of 25 cm from A.

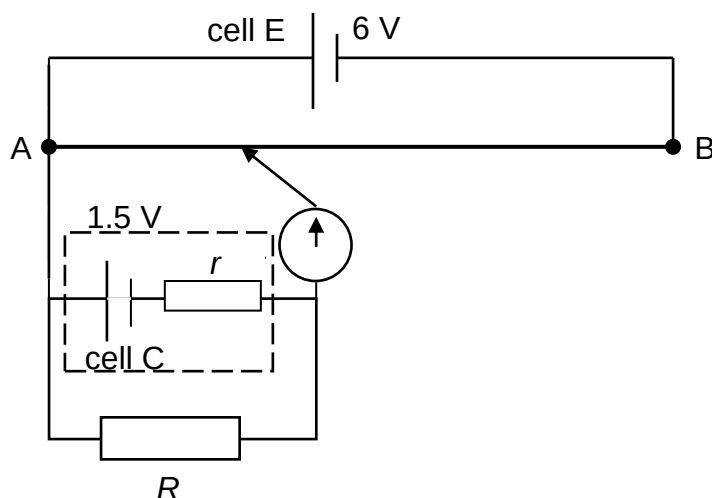


Fig. 6.3

Determine the internal resistance of cell C.

internal resistance = Ω [3]

- (ii) **Solution:**

$$\frac{V_{AP}}{V_E} = \frac{25}{150}$$

$$V_{AP} = \frac{25}{150} \times 6 = 1.0\text{ V}$$

$$V_{AP} = \frac{R}{R+r} V_C$$

$$1.0 = \frac{2}{2+r} (1.5)$$

$$r = 1.0\ \Omega$$

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- (iii) After prolonged use, the internal resistance of cell E may increase and become significant.

State and explain how the balance length mentioned in part (b)(ii) will change.

				[2]
		(iii)	Solution: As internal resistance of cell E increases, its terminal p.d. decreases as $V = E - Ir$. $\frac{V_C}{V} = \frac{I_{AP}}{150}$ Now that V decreases while V_C remains constant, $I_{AP} = \frac{V_C}{V} \times 150$ increases. Therefore the balance length AP increases.	1 1

